AI Contribution Statement

Corporate Finance, Working with AI (WAI) project | Titan Company Ltd | Tanish Tyagi, Roll M56D-26, Section D | IIM Ranchi, Term II

1. Tools used

Tool	Role in the project
Claude (Anthropic), agentic mode with file, shell and web tools	Research assistant and coding assistant: proposed the shock list and the analysis design, searched for sources, wrote the Python and Excel-formula code, drafted the report text, built the dashboard, and ran checks.
Python (pandas, numpy, statsmodels, matplotlib, reportlab, openpyxl)	Independent re-computation of the CAPM regressions, figures for the report, PDF and workbook generation.
LibreOffice (headless)	Recalculation of the workbook so all formulas hold cached values and can be audited without re-running code.
Microsoft Excel-compatible formulas (SLOPE, STEYX, TDIST, SUMPRODUCT)	The analysis itself is live formulas in the workbook, so every number can be traced to a cell.

2. What the AI was used for, by task

WAI task	AI contribution	Author contribution
1. Shock identification	Suggested a long-list of candidate events and the economic channel for each; helped draft the justification.	Chose the final eight shocks and the three featured ones; judged relevance to Titan.
2a/2b. Prices, returns, CAPM beta	Wrote the formulas and Python cross-check; produced the beta plots.	Defined the windows (six months pre and post), the risk-free proxy and the market index; reviewed the outputs.
2c. Financial policy	Extracted line items from the annual reports into a structured sheet; built the ratio, WACC, ROIC and EVA calculations.	Specified the ratios required by the brief and the capex proxy; spot-checked extracted figures against the reports.
3. Value created or destroyed	Drafted the analysis using TSR vs CAPM-required return and ROIC vs WACC; linked to theory.	Reviewed the argument and the reading of each year.
4. CFO recommendations and strategy	Built the five-year scenarios and the stress tests; drafted the recommendations.	Decided which recommendations to adopt; reviewed that they follow from the evidence.
5. References	Suggested the theory references.	Confirmed that each reference is real and is cited for what it says.

3. How the output was verified

- **Two independent calculations.** All eight pre/post betas were recomputed in Python (statsmodels). Excel and Python agree to the last decimal (maximum absolute difference 0.0). A pooled dummy-variable regression gives the same change in beta as the two separate regressions.
- **Seventeen automated integrity tests** in the workbook (Checks sheet), all passing: data completeness and ordering, no implausible daily moves, balance-sheet identity, finance cost, PAT and PBT arithmetic, gold-on-loan and dividend reconciliations to the cash-flow statement, event-window mapping and estimation-window separation.
- **Robustness.** The change in beta is re-estimated under nine specifications (3, 6 and 12-month windows, excluding days around the event, three risk-free proxies, weekly data). Conclusions are reported with the number of specifications that agree, so a result that is fragile is labelled as such.
- **Source tracing.** Financial statement inputs are consolidated figures from the Titan annual reports FY2019-20 to FY2025-26 and are marked as inputs (blue) in the workbook; derived cells are formulas. Market data are the NSE files supplied for the project.
- **Visual and numerical review of every deliverable** (workbook charts, report pages, dashboard in light and dark mode and at phone width) and a check that the dashboard, report and workbook show the same numbers.

4. Limits of the AI-assisted work, disclosed

- The equity risk premium (6%) and the 5x interest-cover floor are assumptions, not data. The NIFTY 500 file is a price index (no dividends), so market returns are slightly understated; Titan's own return includes dividends.
- FY2020 covers nine months of the window and FY2027 has no audited accounts. Some shock windows overlap (for example the two 2022 shocks), so their betas are not independent.
- The counterfactuals and five-year scenarios are illustrative model runs, not forecasts. The recommended policy still breaches the cover floor in two of the stress years; the report says so.
- AI-extracted annual-report figures can contain transcription errors; the reconciliation tests above catch arithmetic inconsistencies but not a consistent misreading, which is why a manual spot-check is listed below.

5. Prompt log (representative prompts used to direct the AI)

#	Purpose	Prompt (written as a professional would specify it)
1	Scope	Act as a corporate-finance analyst. Using the attached brief, list candidate macro or sector shocks between 2020 and 2026 that could plausibly change the systematic risk or financial policy of Titan Company. For each, give the date, the economic channel (demand, cost, rates, regulation, FX) and the evidence you would test it with. Do not include events without a dated trigger.
2	Beta design	Estimate CAPM beta for Titan against the NIFTY 500 using daily excess returns, six months before and six months after each shock date. State the window, the risk-free proxy and the standard errors. Test whether post-shock beta differs from pre-shock beta, and re-estimate under alternative windows to show whether the conclusion is robust.
3	Financial data	Extract consolidated balance sheet, P&L and cash-flow items for FY2019 to FY2026 from the attached annual reports into one structured sheet. Flag every item you could not find. Reconcile the balance sheet and the cash-flow statement before using the data.
4	Policy ratios	Using the extracted financials, compute gross fixed assets to total assets (capex proxy), debt to equity under three definitions of debt, interest cover (EBIT over finance cost), dividends, buybacks and cash, with formulas rather than pasted values.
5	Value	Assess whether the shocks created or destroyed shareholder value. Compare total shareholder return with the CAPM-required return, and ROIC with WACC, by year. Explain any gap using trade-off theory, pecking order and signalling, and say where the evidence does not support the theory.
6	Strategy	Build a five-year model under the current policy and under a recommended policy, each with a base case and a stress case (higher rates, weaker demand). Report interest cover, D/E, ROIC and EVA, and state which guardrails are breached and when.
7	Review	Review the workbook as an auditor would: look for hard-coded numbers in formula cells, inconsistent units, circular references, results that disagree between sheets, and any claim in the report that the data do not support. List each issue with its cell reference.

Author sign-off checklist (to be completed before submission)

[ ] I re-checked at least five financial-statement inputs against the annual reports.

[ ] I re-ran or inspected one beta regression independently and it matches the workbook.

[ ] I read the report and dashboard and I can defend each conclusion in them.

Signature: _____ Date: _____